

Ring-LWE and Ideal Lattice Cryptography _v2

Ring-LWE 與理想格密碼學

Algebraic Number Theory Meets Post-Quantum Security | 代數數論與後量子安全的交匯

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研究動機 (Motivation)

Classical public-key cryptography is provably broken by quantum computation. The two pillars of contemporary public-key infrastructure are RSA (security based on the integer factoring problem) and ECC (security based on the elliptic-curve discrete logarithm problem). **Shor's algorithm** (P. Shor, FOCS 1994) solves both in *quantum polynomial time* $\tilde{O}((\log N)^3)$, whereas the best known classical algorithms (GNFS, Pollard rho) require sub-exponential time $\exp(\tilde{O}((\log N)^{1/3}))$. Once a sufficiently large fault-tolerant quantum computer exists, RSA, Diffie–Hellman, and ECC all collapse simultaneously.

傳統公鑰系統 RSA 與 ECC 之安全性分別建基於**整數分解與橢圓曲線離散對數**。Shor 演算法 (Shor, 1994) 將兩者於量子電腦上同時化為多項式時間 $\tilde{O}((\log N)^3)$ ，較古典最佳演算法 GNFS 之亞指數複雜度顯著加速。一旦容錯量子計算機問世，整個公鑰基礎設施將同時失守。

Lattice-based cryptography provides the leading post-quantum alternative: its security reduces, in the worst case, to geometric problems on lattices (*shortest vector*, *closest vector*) for which no quantum polynomial-time algorithm is known. **Ring-LWE** (Lyubashevsky–Peikert–Regev 2010) further exploits the algebraic structure of number rings to reduce the public key from $O(n^2)$ to $O(n)$ and the multiplication cost from $O(n^2)$ to $O(n \log n)$ via the number-theoretic transform.

格密碼學的安全性可歸約至格上最差情形的 SVP/CVP，迄今無已知量子多項式時間演算法。Ring-LWE 進一步以數環的代數結構，將公鑰自 $O(n^2)$ 壓縮至 $O(n)$ 、乘法自 $O(n^2)$ 加速至 $O(n \log n)$ ，兼得可證明安全與實用效率。

報告架構 (四步驟) (Four-Step Structure)

Step 1 — Lattices and Geometry of Numbers (格與幾何數論)

A **lattice** of rank n is a discrete subgroup $\mathcal{L} = \bigoplus_{i=1}^n \mathbb{Z} b_i \subset \mathbb{R}^n$ generated by linearly independent $b_1, \dots, b_n \in \mathbb{R}^n$. The basis matrix $B = [b_1 \mid \dots \mid b_n]$ defines the *covolume* $\det(\mathcal{L}) = |\det B|$, an invariant of $\mathcal{L}$. The successive minima are

$$\lambda_i(\mathcal{L}) := \inf\{r > 0 : \dim \text{span}(\mathcal{L} \cap \overline{B(0, r)}) \geq i\}, \quad i = 1, \dots, n.$$

The central hard problem is SVP_γ : find $v \in \mathcal{L} \setminus \{0\}$ with $\|v\| \leq \gamma \lambda_1$. (Variants: GapSVP_γ , BDD_d .) The best classical and quantum algorithms at small γ require time $2^{\Theta(n)}$.

Minkowski's first theorem. For any centrally symmetric convex body $S \subset \mathbb{R}^n$ with $\text{vol}(S) > 2^n \det(\mathcal{L})$, S contains a nonzero lattice point.

Deriving the SVP upper bound. Apply Minkowski's theorem to the centered cube $C_r := [-r, r]^n$, whose volume is $\text{vol}(C_r) = (2r)^n$. The hypothesis $\text{vol}(C_r) > 2^n \det(\mathcal{L})$ becomes

$$(2r)^n > 2^n \det(\mathcal{L}) \iff r > \det(\mathcal{L})^{1/n}.$$

Whenever this holds, C_r contains a nonzero $v \in \mathcal{L}$. Since $v \in C_r$ means $\|v\|_\infty \leq r$, and the standard norm inequality $\|v\|_2 \leq \sqrt{n} \|v\|_\infty$ holds for any $v \in \mathbb{R}^n$, we obtain $\|v\|_2 \leq \sqrt{n} r$. Letting $r \searrow \det(\mathcal{L})^{1/n}$,

$$\lambda_1(\mathcal{L}) \leq \sqrt{n} \cdot \det(\mathcal{L})^{1/n}.$$

This bound is sharpened via the Hermite constant $\gamma_n \leq \frac{2n}{\pi e} + o(n)$.

格 $\mathcal{L} \subset \mathbb{R}^n$ 之核心不變量為覆積 $\det(\mathcal{L})$ 與相繼極小 λ_i ， SVP_γ 為主要困難問題。由 **Minkowski 定理** 推導 **SVP 上界**：取中心對稱立方體 $C_r = [-r, r]^n$ ，體積 $(2r)^n$ ；當 $r > \det(\mathcal{L})^{1/n}$ 時體積超過 $2^n \det(\mathcal{L})$ ，故 C_r 含非零格點 v ，由 $\|v\|_\infty \leq r$ 與 $\|v\|_2 \leq \sqrt{n} \|v\|_\infty$ 得 $\|v\|_2 \leq \sqrt{n} r$ ；令 $r \searrow \det(\mathcal{L})^{1/n}$ 即得 $\lambda_1 \leq \sqrt{n} \det(\mathcal{L})^{1/n}$ 。此即 **Course Objective 5** 之核心。

Step 2 — Cyclotomic Fields and Ideal Lattices (分圓域與理想格)

General n -th cyclotomic field. For any integer $n \geq 1$, let $\zeta_n = e^{2\pi i/n}$ be a primitive n -th root

of unity. The n -th **cyclotomic polynomial** is

$$\Phi_n(x) = \prod_{\substack{1 \leq k \leq n \\ \gcd(k, n) = 1}} (x - \zeta_n^k) \in \mathbb{Z}[x],$$

which is monic, irreducible over $\mathbb{Q}$, of degree $\varphi(n)$ (Euler's totient). Small examples:

$$\Phi_1 = x - 1, \quad \Phi_2 = x + 1, \quad \Phi_3 = x^2 + x + 1, \quad \Phi_4 = x^2 + 1, \quad \Phi_6 = x^2 - x + 1, \quad \Phi_8 = x^4 + 1.$$

The n -th **cyclotomic field** $K = \mathbb{Q}(\zeta_n)$ has degree $[K : \mathbb{Q}] = \varphi(n)$, and the ring of integers is $\mathcal{O}_K = \mathbb{Z}[\zeta_n]$ —a Dedekind domain (**Obj. 1, 2**); every nonzero ideal admits a unique factorisation $\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}$.

Specialisation: power-of-two cyclotomic ($m = 2^k, k \geq 2$). For Ring-LWE we use $m = 2^k$, since then $\Phi_m(x) = x^{2^{k-1}} + 1$ takes the simplest binomial form. Following the LPR convention, henceforth m denotes the cyclotomic index and $n := \varphi(m) = 2^{k-1}$ the field degree.

Splitting of primes (Obj. 3). For a rational prime $p \nmid m$, let f be the order of p modulo m . Then $p\mathcal{O}_K = \mathfrak{p}_1 \cdots \mathfrak{p}_g$ splits into $g = n/f$ distinct primes of inertia degree f . For $p = 2$ (the unique ramified prime when $m = 2^k$), $2\mathcal{O}_K = (1 - \zeta_m)^n$ is *totally ramified*.

Trace, norm, discriminant (Obj. 4). For $\alpha \in K$, $\text{Tr}_{K/\mathbb{Q}}(\alpha) = \sum_i \sigma_i(\alpha)$ and $N_{K/\mathbb{Q}}(\alpha) = \prod_i \sigma_i(\alpha)$. The *different* $\mathfrak{d}_{K/\mathbb{Q}}^{-1} = \{\alpha \in K : \text{Tr}(\alpha\mathcal{O}_K) \subseteq \mathbb{Z}\}$ satisfies $\Delta_K = N_{K/\mathbb{Q}}(\mathfrak{d}_{K/\mathbb{Q}})$. For $m = 2^k$: $|\Delta_K| = 2^{n(k-1)}$.

Canonical (Minkowski) embedding. Cyclotomic fields are totally complex ($r_1 = 0, r_2 = n/2$), so

$$\sigma: K \hookrightarrow K_{\mathbb{R}} := K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{C}^{n/2} \cong \mathbb{R}^n, \quad \alpha \mapsto (\sqrt{2} \Re \sigma_1(\alpha), \sqrt{2} \Im \sigma_1(\alpha), \dots).$$

Under this Minkowski normalisation, every nonzero ideal $\mathfrak{a} \subseteq \mathcal{O}_K$ becomes a full-rank **ideal lattice** with covolume $\det(\sigma(\mathfrak{a})) = N(\mathfrak{a}) \sqrt{|\Delta_K|}$.

Worked example ($m = 8, n = 4$). $K = \mathbb{Q}(\zeta_8)$, $\Phi_8(x) = x^4 + 1$, $|\Delta_K| = 2^{4 \cdot 2} = 256$, $\sqrt{|\Delta_K|} = 16$. For $q = 17 \equiv 1 \pmod{8}$: $f = 1$, so $17\mathcal{O}_K = \mathfrak{q}_1 \mathfrak{q}_2 \mathfrak{q}_3 \mathfrak{q}_4$ splits completely—foreshadowing the NTT factorisation of R_{17} in Step 3.

一般 n -th cyclotomic field：取 $\zeta_n = e^{2\pi i/n}$ ， $\Phi_n(x)$ 為其極小多項式，係數為整數、不可約、次數 $\varphi(n)$ (Euler totient)；列舉 Φ_1 至 Φ_8 為小例。整數環 $\mathcal{O}_K = \mathbb{Z}[\zeta_n]$ 為 Dedekind domain，貫穿 **Obj. 1–4**。特殊化至 $m = 2^k$ ： $\Phi_m(x) = x^n + 1$ 形式最簡，Ring-LWE 實作首選；以下 m 代分圓指標、 $n := \varphi(m) = 2^{k-1}$ 代域度。Minkowski 嵌入下理想格之覆

積為 $N(\mathfrak{a})\sqrt{|\Delta_K|}$ ，此即把代數結構翻譯為幾何結構的關鍵公式。 $m = 8$ 時 $|\Delta_K| = 256$ 、 $q = 17$ 完全分裂為四個質理想，預示 Step 3 的 NTT。

Step 3 — LWE and Ring-LWE (LWE 與其環版本)

Intuition. LWE asks: *given a noisy linear function, recover the secret coefficients.* Given samples (a_i, b_i) with $b_i \approx \langle a_i, s \rangle \pmod{q}$ —each b_i perturbed by a small random error e_i — the goal is to recover s . Without noise ($e_i = 0$), Gaussian elimination solves this in polynomial time; with even small noise, no efficient algorithm is known.

LWE (Regev, STOC 2005 / JACM 2009): fix n, q and an error distribution χ on $\mathbb{Z}_q$ (typically discrete Gaussian of width αq with $\alpha < 1/\sqrt{n}$). For fixed $s \in \mathbb{Z}_q^n$, an LWE sample is

$$(a, \langle a, s \rangle + e \bmod q) \in \mathbb{Z}_q^n \times \mathbb{Z}_q, \quad a \leftarrow U(\mathbb{Z}_q^n), \quad e \leftarrow \chi.$$

Regev’s quantum reduction: $\text{LWE}_{n,q,\chi} \leftarrow \text{GapSVP}_\gamma, \text{SIVP}_\gamma$ on arbitrary lattices, $\gamma = \tilde{O}(n/\alpha)$.

Drawback: the public key $A \in \mathbb{Z}_q^{n \times m}$ has size $O(n^2)$, with $O(n^2)$ multiplication.

Lift to the ring of algebraic integers —Ring-LWE (Lyubashevsky–Peikert–Regev, EUROCRYPT 2010 / JACM 2013): replace integer vectors $\mathbb{Z}^n$ by elements of the cyclotomic ring of integers

$$R := \mathcal{O}_K = \mathbb{Z}[x]/\Phi_m(x) = \mathbb{Z}[x]/(x^n + 1), \quad R_q := R/qR.$$

Both secret s and samples $(a, a \cdot s + e)$ now live in R_q (resp. $R_q \times R_q^\vee$, where $R^\vee := \mathfrak{d}_{K/\mathbb{Q}}^{-1}$ is the codifferent, **Obj. 4**). **Reduction:** Ring-LWE $\leftarrow$ approx-SVP on *ideal lattices* in R (quantum, polynomial-time). For power-of-two cyclotomics, $R^\vee = \frac{1}{n}R$ collapses to a scalar dilation, giving the *simplified* Ring-LWE used in implementations.

The role of prime splitting (Obj. 3). Choose q prime with $q \equiv 1 \pmod{m}$. Then $\Phi_m(x)$ has n distinct roots in $\mathbb{Z}_q$:

$$\Phi_m(x) \equiv \prod_{i=1}^n (x - \omega_i) \pmod{q}, \quad R_q \cong \prod_{i=1}^n \mathbb{Z}_q \text{ (CRT)}.$$

This isomorphism is precisely the **Number-Theoretic Transform (NTT)**, computable in $O(n \log n)$. Multiplication in R_q becomes pointwise after NTT, mirroring convolution-via-FFT; geometrically, $q\mathcal{O}_K$ splits completely into n distinct primes.

Negacyclic convolution & key-size saving. Since $\Phi_m(x) = x^n + 1$, $x^n \equiv -1$ in R_q , giving $(a * b)_k = \sum_{i+j=k} a_i b_j - \sum_{i+j=k+n} a_i b_j$ for $0 \leq k < n$. The Ring-LWE public key has total size $O(n)$ —a factor- n saving over LWE.

直觀解釋：LWE 是帶雜訊的線性方程組求解。給定樣本 (a_i, b_i) 滿足 $b_i \approx \langle a_i, s \rangle \pmod{q}$ (每筆有小量隨機誤差 e_i)，目標還原秘密 s 。若無雜訊 ($e_i = 0$)，高斯消去法多項式時間可解；加上小雜訊後問題變為指數困難。LWE 之安全性歸約至任意格上的 worst-case SVP/SIVP (量子歸約)。**提升至代數整數環：**Ring-LWE 將 $\mathbb{Z}^n$ 換為分圓整數環 $R = \mathcal{O}_K$ ，secret 與 sample 皆為 ring elements；安全性歸約至理想格上的 approx-SVP。質理想完全分裂 (**Obj. 3**) 對應 NTT 之 CRT 同構，公鑰自 $O(n^2)$ 縮為 $O(n)$ 、乘法自 $O(n^2)$ 加速至 $O(n \log n)$ 。

Step 4 — Applications (應用)

日常加密連線 (HTTPS、行動銀行、訊息、VPN) 皆仰賴 RSA/ECC，量子電腦出現將同時失效；且 **今日攔截、未來解密** 的威脅已存在。NIST 於 2024 年 8 月公布首批後量子密碼標準 (FIPS 203 與 FIPS 204)，分別作為金鑰交換與數位簽章的後量子替代方案，皆以 Ring-LWE 為數學基礎。

主要參考文獻：O. Regev, *J. ACM* **56**(6), 2009. Lyubashevsky–Peikert–Regev, *J. ACM* **60**(6), 2013. C. Peikert, *Found. Trends TCS* **10**(4), 2016. NIST FIPS 203/204, 2024. D. A. Marcus, *Number Fields*, Ch. 1–5.